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powerful flight muscle which arises from the sternum. Contraction of these muscles lower the wings in flight. Pectoralis minor (subclavius) is small and elongated muscle which elevates the wings during flight. Besides the pectoralis, the small **coracobrachialis** muscle also helps to pull the wings down and to rotate wings during flight.

Digestive System

The long coiled alimentary canal consist of buccal cavity, pharynx, oesophagus, crop, stomach, small intestine and Large intestine. (Figure 4.29)

Mouth is covered by a toothless, horny, upper and lower **beaks**. Behind the mouth, there is a wide buccal cavity. In the floor of the **buccal cavity**, a large, narrow, horny **tongue** is present with scanty **sensory papillae** and numerous **mucus glands**. Buccal cavity leads into the pharynx followed by the oesophagus, which enlarges to form a thin walled, bilobed elastic sac, the **crop**. The crop serves as a food reservoir. Beyond the crop the **oesophagus** enters the **stomach** which is differentiated into anterior glandular **proventriculus** and a posterior muscular **ventriculus** or **gizzard**. The proventriculus has a mucus lining which secretes the gastric juice. The walls of the gizzard is thick, muscular and has many tubular glands. The cavity of the gizzard contains grit or small pebbles called **gastroliths** that are swallowed by the bird. These stones helps the bird in grinding the food. The gizzard leads to a **small intestine** which consists of a 'U' shaped **duodenum** and **ileum**. The **pancreas** lies between the two limbs of the duodenum and receives three ducts from the pancreas and two bile ducts from the **liver**. The inner lining of the ileum contains numerous **villi** which helps in absorption. The ileum continues into the large intestine, which is short and is differentiated into rectum and cloaca. A pair of small blind pouches called **rectal caeca** is present at the junction of the ileum and rectum. The rectum leads into the **cloaca** which is divided into the anterior **coprodaeum** into which the rectum opens, the middle **urodaeum** into which the **urinigenital** ducts open, and the posterior **vestibule** or **proctodaeum**, which opens to the outside by the cloacal aperture.

Buccal glands, salivary glands, gastric glands, liver, pancreas and intestinal glands are the **digestive glands** which enhance the process of digestion in pigeon. There is **no gall bladder** in the pigeon though present in many other birds. Pigeons produce 'milk', a cheesy and nourishing secretion, from both the sexes. It is formed by the degeneration of the epithelial cells lining the crop. It is regurgitated and fed to the young birds.

The pigeon feeds on grains. As birds have no teeth, the food swallowed by it passes through the **gullet** or **oesophagus** into the **crop** where it is stored. There are mucous glands in the crop; food is softened by being mixed with the **mucus** and the secretion of the **buccal glands,** aided by the warmth of the body. The food then enters the **stomach,** where it is digested by **gastric juices** secreted in the **proventriculus**; the food is also **crushed** in the **gizzard** with aid of gastroliths. The food is thus reduced to smaller particles and the partly digested food passes into the **intestine** where it is